Floating-Strike Lookback Options

Closed Form, Exact-Bridge and Time-Stepping Monte Carlo, Greeks and a Code Review

Mohammed-Yassine Barnicha

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Abstract

We price European floating-strike lookback options under Black–Scholes with a continuous dividend, at issue and during their life. Two Monte Carlo engines written in C++ are studied: one draws the terminal value and the running extremum exactly, from the law of the Brownian bridge, in a single step; the other simulates paths on a time grid and corrects the discrete-monitoring bias with the Broadie–Glasserman–Kou shift. We derive the law of the running extremum, the Goldman–Sosin–Gatto closed form, and the scaling properties of the price: at issue the delta equals V/S, and the issue price has no curvature, while a contract already held keeps a positive gamma $g''(1)/S$, a distinction the original Greeks missed. We add an exact bridge on each time step, which removes the discretisation bias whatever the number of steps, quantify the bias of each scheme, and list the defects found in the C++ code. Everything is reproduced by the interactive page `https://yassinebarnicha.github.io/options-lookback/`, whose JavaScript engines match the C++ to 10^{-15}.

Contents

1 Introduction **2**

2 Contract and model **2**

3 The law of the running extremum **2**

4 The closed form **3**

5 Homogeneity and the Greeks **3**
5.1 Scaling . . . 3
5.2 Monte Carlo Greeks with common random numbers . . . 4

6 The Monte Carlo engines **4**
6.1 Engine A: exact bridge in one step . . . 4
6.2 Engine B: time stepping . . . 4
6.3 Added: the exact bridge on each step . . . 5

7 Numerical study **5**
7.1 Discretisation bias . . . 5
7.2 Convergence and antithetic variates . . . 5
7.3 The sampler against the exact law . . . 6
7.4 Reference values . . . 6

8 Code review **7**

9 Validation of the port **7**

10 Conclusion 7

1 Introduction

A floating-strike lookback call pays $S_T - \min_{[0,T]} S$: the holder buys at the lowest price of the period. The put pays $\max_{[0,T]} S - S_T$. Both payoffs are path-dependent but only through one running extremum, which makes them a good test bed for Monte Carlo methods: the exact price is known in closed form, and the error of a discretely monitored simulation can be measured exactly.

The project contains two independent C++ implementations, wrapped in a common DLL and called from Excel/VBA:

- *engine A* (Barnicha) draws $(S_T, \text{extremum})$ exactly in one step by the Brownian bridge;
- *engine B* (Chivet) simulates a time grid and applies the BGK correction to the discrete extremum.

This report gives the theory behind both (sections 3, 4 and 6), the scaling properties that determine the Greeks (section 5), a numerical study (section 7) and the code review (section 8).

2 Contract and model

Under the risk-neutral measure, $S_t = S\exp(\mu t + \sigma W_t)$ with $\mu = b - \sigma^2/2$, where $b = r - q$ is the carry (r rate, q continuous dividend yield). Write $X_t = \ln(S_t/S)$, a Brownian motion with drift μ and volatility σ, and $M_T = \max_{[0,T]} X$, $m_T = \min_{[0,T]} X$. For a contract issued in the past, let $M \geq S$ (put) or $m \leq S$ (call) be the extremum already observed. The payoffs at T (time to maturity from now) are

$$C_T = S_T - \min\left(m,\ S\,e^{m_T}\right), \qquad P_T = \max\left(M,\ S\,e^{M_T}\right) - S_T, \tag{1}$$

with $m = M = S$ at issue. Prices are $V = e^{-rT}\mathbb{E}[\,\cdot\,]$; since $\mathbb{E}[e^{-rT}S_T] = Se^{-qT}$,

$$V^{\text{put}} = e^{-rT}\mathbb{E}\big[\max(M, Se^{M_T})\big] - S\,e^{-qT}, \qquad V^{\text{call}} = S\,e^{-qT} - e^{-rT}\mathbb{E}\big[\min(m, Se^{m_T})\big]. \tag{2}$$

3 The law of the running extremum

Proposition 1 (Joint law)**.** *For $y \geq \max(0, x)$,*

$$\mathbb{P}(M_T \geq y,\ X_T \in \mathrm{d}x) = \frac{1}{\sigma\sqrt{T}}\,\varphi\!\left(\frac{2y - x + \mu T}{\sigma\sqrt{T}}\right) e^{2\mu y/\sigma^2}\,\mathrm{d}x. \tag{3}$$

Proof. Without drift ($\mu = 0$, $\sigma = 1$), the reflection principle states that the path reflected after the first passage at y is again a Brownian motion, so $\mathbb{P}(M_T \geq y, W_T \in \mathrm{d}x) = \mathbb{P}(W_T \in \mathrm{d}(2y - x))$. The drift is added by the Girsanov density $\exp(\mu x/\sigma^2 - \mu^2 T/2\sigma^2)$, which depends on the end point only; combining the exponents gives the formula. □

Integrating in x gives the marginal law

$$\mathbb{P}(M_T \leq y) = \Phi\!\left(\frac{y - \mu T}{\sigma\sqrt{T}}\right) - e^{2\mu y/\sigma^2}\,\Phi\!\left(\frac{-y - \mu T}{\sigma\sqrt{T}}\right), \qquad y \geq 0, \tag{4}$$

and symmetrically $\mathbb{P}(m_T \geq y) = \Phi\big(\frac{-y+\mu T}{\sigma\sqrt{T}}\big) - e^{2\mu y/\sigma^2}\Phi\big(\frac{y+\mu T}{\sigma\sqrt{T}}\big)$ for $y \leq 0$. Dividing the joint law by the density of X_T removes the drift:

Proposition 2 (Brownian bridge)**.** *Conditionally on $X_0 = x_0$ and $X_\tau = x_1$, for $y \geq \max(x_0, x_1)$,*

$$\mathbb{P}\Big(\max_{[0,\tau]} X \geq y \,\Big|\, x_0, x_1\Big) = \exp\!\left(-\frac{2(y - x_0)(y - x_1)}{\sigma^2\tau}\right), \tag{5}$$

and the same holds for the minimum with $y \leq \min(x_0, x_1)$.

The law does not depend on μ: once both end points are known, the drift carries no information about the path.

4 The closed form

Proposition 3 (Goldman, Sosin and Gatto; general form)**.** *With* $b_1 = \dfrac{\ln(S/M) + (b+\sigma^2/2)T}{\sigma\sqrt{T}}$ *and* $b_2 = b_1 - \sigma\sqrt{T}$,

$$V^{\text{put}} = -S\,e^{(b-r)T}\Phi(-b_1) + Me^{-rT}\Phi(-b_2) + S\,e^{-rT}\frac{\sigma^2}{2b}\left[-\Big(\frac{S}{M}\Big)^{-2b/\sigma^2}\Phi\Big(b_1 - \frac{2b}{\sigma}\sqrt{T}\Big) + e^{bT}\Phi(b_1)\right], \tag{6}$$

and, with $a_1 = \dfrac{\ln(S/m) + (b+\sigma^2/2)T}{\sigma\sqrt{T}}$, $a_2 = a_1 - \sigma\sqrt{T}$,

$$V^{\text{call}} = S\,e^{(b-r)T}\Phi(a_1) - me^{-rT}\Phi(a_2) + S\,e^{-rT}\frac{\sigma^2}{2b}\left[\Big(\frac{S}{m}\Big)^{-2b/\sigma^2}\Phi\Big(-a_1 + \frac{2b}{\sigma}\sqrt{T}\Big) - e^{bT}\Phi(-a_1)\right]. \tag{7}$$

Sketch. In (2), with $k = \ln(M/S) \geq 0$, $\mathbb{E}[\max(M, Se^{M_T})] = M\,\mathbb{P}(M_T \leq k) + S\int_k^\infty e^y\,\mathrm{d}F(y)$, where F is (4). The first term gives $Me^{-rT}\Phi(-b_2)$ after discounting. In the integral, e^y times the Gaussian part of $\mathrm{d}F$ is a Gaussian of shifted mean (completing the square), which produces the $\Phi(b_1)$ terms, and e^y times the reflected part, $e^{(1+2\mu/\sigma^2)y}$ times a Gaussian, produces the power $(S/M)^{-2b/\sigma^2}$ and the factor $\sigma^2/2b$ through $1 + 2\mu/\sigma^2 = 2b/\sigma^2$. □

Remark 1 (Zero carry). Each bracket vanishes when $b \to 0$ and the limit is finite. The general formula is evaluated at $b = \pm 10^{-6}$ and averaged when $|b| < 10^{-6}$, an error of order 10^{-12}; engine A codes the exact limit for $q = 0$, $r \to 0$; engine B divides by $2r$ and is undefined at $r = 0$.

On the reference case ($S = 100$, $r = 5\,\%$, $q = 0$, $\sigma = 20\,\%$, $T = 1$) the put is worth 14.290568 and the call 17.216802.

5 Homogeneity and the Greeks

5.1 Scaling

The dynamics is multiplicative and the payoffs are positively homogeneous in $(S, \text{extremum})$, hence

$$V(\lambda S, \lambda m) = \lambda\,V(S,m) \quad (\lambda > 0), \qquad V(S,m) = S\,g(m/S), \tag{8}$$

for a function g depending on (r, q, σ, T). By Euler's theorem $V = S\,\partial_S V + m\,\partial_m V$.

Lemma 4. *At* $m = S$, $\partial_m V = 0$ *(one-sided derivative).*

Proof. For the call, $\partial_m \min(m, Se^{m_T}) = \mathbf{1}\{m < Se^{m_T}\}$, so $\partial_m V = -e^{-rT}\mathbb{P}(\min_{[0,T]} S_t > m)$. As $m \uparrow S$ this probability tends to $\mathbb{P}(m_T > 0) = 0$: a Brownian motion goes below its starting point immediately. The put is symmetric. □

Proposition 5 (Greeks at issue)**.** *With* $z = m/S$, $\Delta = \partial_S V = g(z) - z\,g'(z)$ *and* $\Gamma = \partial_{SS} V = z^2 g''(z)/S$. *At issue* ($z = 1$)*:*

(i) $\Delta = g(1) = V/S$*, the same for the issue price* $S \mapsto V(S,S)$ *and for the contract held (proposition 4);*

(ii) the issue price $V(S,S) = S\,g(1)$ *is linear in* S*: its second derivative is* 0*;*

(iii) the contract held, whose extremum stays at m *when the spot moves, has* $\Gamma = g''(1)/S > 0$*;*

(iv) vega, rho and theta are proportional to S*.*

Point (iii) matters for hedging: right after issue the extremum is a fixed number in the contract, not a function of the spot. On the reference case the held gamma is 0.0312 for the put and 0.0455 for the call, while the issue-price gamma is zero. Both C++ engines bump the spot *and* the starting extremum, so they report the gamma of (ii), not of (iii). The page computes (iii) by fixing the extremum (seasoned mode); fig. 1 shows both.

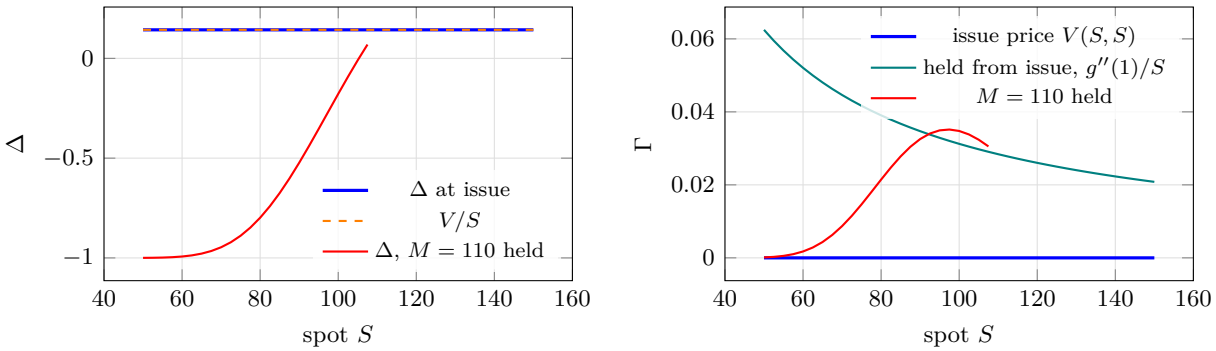


Figure 1: Put, reference parameters. Left: at issue $\Delta = V/S = 0.1429$ for every spot; with a maximum $M = 110$ already observed, Δ varies. Right: the issue price has zero gamma, the contract held does not.

5.2 Monte Carlo Greeks with common random numbers

Both engines compute Greeks by bump and revalue on *common random numbers* (CRN): the bumped prices reuse the same draws. For a parameter θ and a payoff $Y(\theta, \omega)$ Lipschitz in θ with constant $L(\omega)$,

$$\operatorname{Var}\left[\tfrac{Y(\theta+h)-Y(\theta-h)}{2h}\right] \le \mathbb{E}[L^2] \quad \text{(CRN)}, \qquad \operatorname{Var}\left[\tfrac{Y(\theta+h)-Y'(\theta-h)}{2h}\right] = \frac{\operatorname{Var} Y}{2h^2} \quad \text{(independent draws)},$$

so CRN keeps the variance bounded as $h \to 0$. At issue each path's payoff is itself homogeneous of degree one in S (the whole path and the starting extremum scale with S). Hence, path by path, the bumped delta equals Y/S and the bumped gamma is exactly zero: engine A returns $\Gamma = -1.1 \cdot 10^{-10} \pm 1.4 \cdot 10^{-12}$ (rounding) and a delta whose standard error is that of the price divided by S.

Theta conventions. Engine A computes $(V(T-h) - V(T))/h$, a one-sided estimate of $-\partial_T V$; engine B computes $+\partial_T V$ by central differences. The page keeps both conventions and offers a centred version of the first.

6 The Monte Carlo engines

6.1 Engine A: exact bridge in one step

Draw $X_T = \mu T + \sigma\sqrt{T}\, Z$, $Z \sim \mathcal{N}(0,1)$, then the extremum by inverting (5) with $x_0 = 0$, $x_1 = X_T$ and a uniform U: solving $\exp(-2y(y - X_T)/\sigma^2 T) = U$ for y,

$$M_T = \tfrac{1}{2}\Big(X_T + \sqrt{X_T^2 - 2\sigma^2 T \ln U}\Big), \qquad m_T = \tfrac{1}{2}\Big(X_T - \sqrt{X_T^2 - 2\sigma^2 T \ln U'}\Big). \tag{9}$$

The pair (X_T, M_T) has exactly the law of proposition 1: the estimator is unbiased with a single step. The antithetic pair $(Z, -Z)$ with the same uniforms reduces the variance by a factor of about 2 on the reference case (fig. 3). U and U' are drawn independently, which is correct for these payoffs (each uses one extremum) but would not give the joint law of (M_T, m_T) needed for a payoff such as $\max - \min$.

6.2 Engine B: time stepping

Paths are simulated on n steps of length $\Delta t = T/n$ and the extremum is read on the grid. The discrete maximum is smaller than the continuous one, and for a Brownian motion the gap is of

order $\sqrt{\Delta t}$ (Asmussen, Glynn and Pitman 1995):

$$\mathbb{E}\Big[\max_{[0,T]} X\Big] - \mathbb{E}\Big[\max_{k\le n} X_{k\Delta t}\Big] = \beta\,\sigma\sqrt{\Delta t} + o(\sqrt{\Delta t}), \qquad \beta = -\frac{\zeta(1/2)}{\sqrt{2\pi}} \approx 0.5826. \tag{10}$$

Broadie, Glasserman and Kou (1997, 1999) turn this into a correction: the discrete extremum is shifted, $\max S \leftarrow e^{\beta\sigma\sqrt{\Delta t}} \max S$ and $\min S \leftarrow e^{-\beta\sigma\sqrt{\Delta t}} \min S$, which removes the first-order term; the residual bias is $O(\Delta t)$. This is what the C++ does.

6.3 Added: the exact bridge on each step

Conditionally on the grid values, the pieces of path on different steps are independent Brownian bridges (Markov property). The continuous maximum is the maximum over the steps of the bridge maxima, each of which can be drawn from (5) with its own uniform:

$$\max_{[0,T]} X = \max_{k} \tfrac{1}{2}\Big(X_k + X_{k+1} + \sqrt{(X_{k+1}-X_k)^2 - 2\sigma^2\Delta t \ln U_k}\Big). \tag{11}$$

The scheme is then exact for continuous monitoring for any n; the grid only matters if the payoff needs intermediate values. Its cost is one uniform and one square root per step.

7 Numerical study

7.1 Discretisation bias

Figure 2 shows the bias of engine B against the closed form, 20 000 paths per point, for the put. Without correction the bias is -1.25 with 100 steps (a 9 % underpricing) and decreases like $\sqrt{\Delta t}$. BGK overcorrects at coarse grids ($+2.2$ with 2 steps, $+0.35$ with 10) and its residual decreases like Δt; it is within the statistical error from about 50 steps. The bridge per step has no visible bias at any n, including $n = 2$.

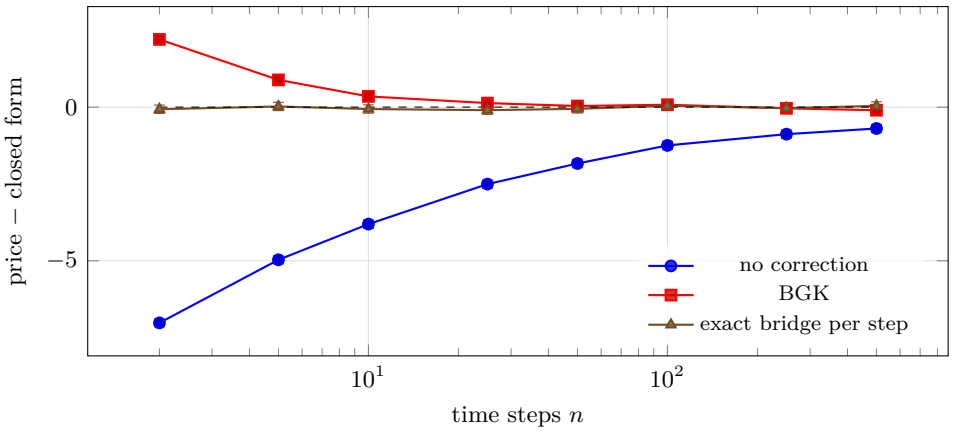


Figure 2: Discretisation bias of the time-stepping engine, put, reference parameters, with 95 % confidence intervals.

7.2 Convergence and antithetic variates

Figure 3 shows engine A against the closed form. Both estimators converge at the Monte Carlo rate $N^{-1/2}$; the antithetic version has a standard error 1.42 times smaller for the same number of normal draws (variance divided by 2.0).

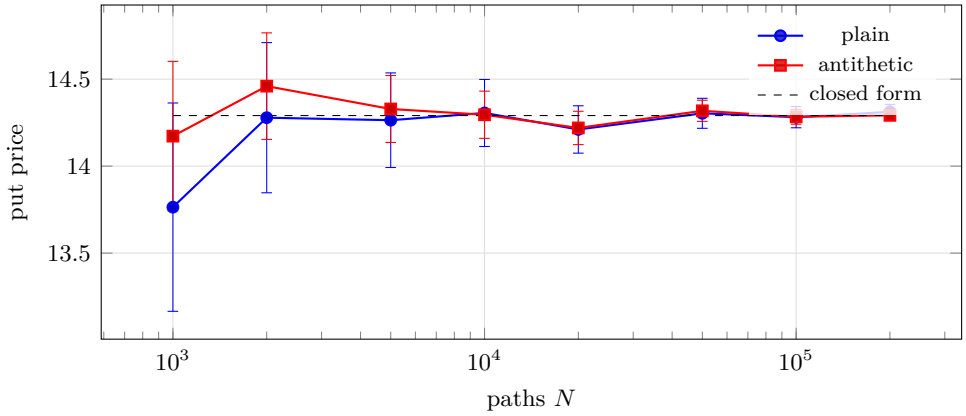


Figure 3: Exact-bridge engine against the closed form, 95 % confidence intervals.

7.3 The sampler against the exact law

Figure 4 compares the maxima drawn by (9) (200 000 paths) with the density from (4): the histogram follows the exact law, including the mass near 0 (paths that never rise much above the start).

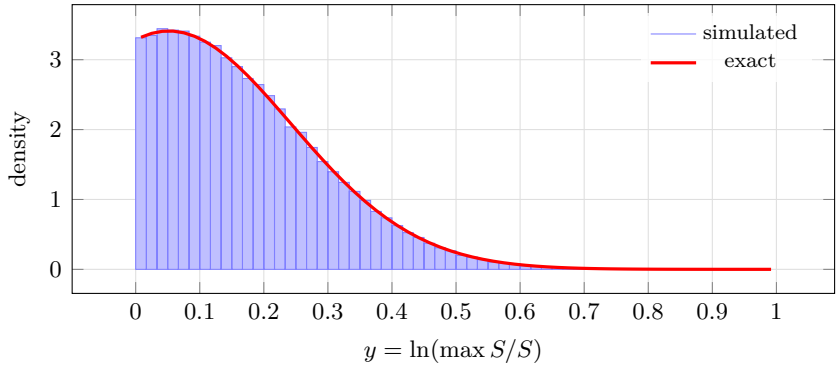


Figure 4: Law of M_T (reference parameters): bridge sampler against (4).

7.4 Reference values

	closed form	engine A (10^4 paths)	engine B, no correction (10^4 paths × 100 steps)
price	14.2906	14.2953 ± 0.1357	12.9073 ± 0.1925
Δ	0.142906	0.142953	0.129073
Γ (issue price)	0	$-1.1 \cdot 10^{-10}$	$1.9 \cdot 10^{-11}$
vega	87.170	87.184	79.460
ρ	−49.379	−49.371	−48.900

Table 1: Put with the default settings of the Excel workbook ($S = 100$, $r = 5\,\%$, $\sigma = 20\,\%$, $T = 1$, seed 42), Monte Carlo prices with 95 % confidence intervals. Engine B with BGK gives 14.2320; with the bridge per step it is unbiased.

8 Code review

1. *The engine A program does not compile*: `Barnicha/app/main.cpp` has `using namespace lookback;` while the namespace is `Barnicha`. It is the executable called by the Excel macros.

2. *Engine B ignores the seed*: `getRandomSeed()` returns the clock, so two identical runs differ.

3. *Engine B, imperfect common random numbers*: `setRandomSeed` reseeds the generator but the normal law keeps the value held in reserve by the polar method, so the bumped prices do not use the same normals. This defeats the purpose of CRN and adds noise to every Greek.

4. *Engine B, standard error of ρ*: the estimate divides by $2hr$ (relative bump), the standard error by $2h$ only.

5. *Engine B, closed-form gamma four times too small*: divided by $(0.0002\,S)^2$ instead of $(0.0001\,S)^2$; invisible at issue, where the issue-price gamma is zero.

6. *Engine B, $r = 0$*: the relative bump on r and the closed form (division by $2r$) are undefined.

7. *Opposite theta conventions* between the two engines, one of them one-sided.

8. *Gamma at issue is not the hedging gamma*: both engines bump the starting extremum with the spot (proposition 5).

9. *Independent extremum draws* in engine A: fine for these payoffs, wrong for a payoff using both extrema.

10. *`out_size` is never set* by the exported functions: the caller must know that there are 48 outputs.

The JavaScript port fixes 2–6 (as options where the original behaviour is kept for comparison) and adds the dividend, the seasoned extremum, the bridge per step, antithetic variates, a centred theta and adjustable bumps.

9 Validation of the port

The generator `std::mt19937` and libstdc++'s `generate_canonical`, uniform and normal (polar Marsaglia) distributions are reproduced bit for bit. With the same seed, engine A returns the prices of the C++ program compiled with g++ to 10^{-15} and engine B those of the C++ simulator (with and without BGK) to 10^{-12}; the closed-form Greeks match the Excel workbook to 10^{-10}. With a dividend or a seasoned extremum, the Monte Carlo recovers the general closed form within its confidence interval.

10 Conclusion

The exact bridge makes the lookback a one-step simulation: unbiased, cheap, and easily extended to any number of monitoring steps by drawing one bridge extremum per step. Time stepping with BGK is a good approximation from about fifty steps but overcorrects on coarse grids, while uncorrected discrete monitoring underprices by several percent. The scaling of the price explains the Greeks: the delta at issue is V/S, and the gamma reported at issue is the curvature of the issue price, which is zero, while the gamma of a contract held is $g''(1)/S$. The review found ten defects in the C++ code, the most important being the uncompilable engine A program and the broken common random numbers of engine B.

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